2.3 The particle model

Science inquiry skills

FOUNDATION | STANDARD | ADVANCED

Processing & Analysing

Four balloons were blown up to different sizes in different rooms of a house. The temperature of each room was different. The balloons are shown in Figure 2.3.1.

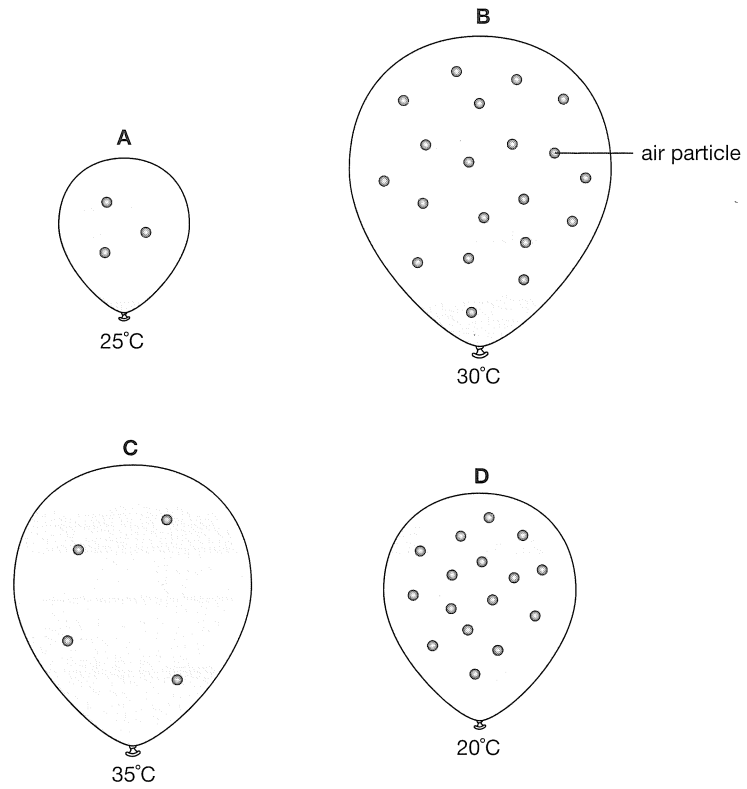


Figure 2.3.1

(1) Identify the balloon (A, B, C or D):

(a) in which the air particles would be moving the fastest _____

(b) in which the air particles would be moving the slowest _____

(c) in which the air particles are furthest apart _____

(d) in which the air particles are closest to each other _____

(e) that would be the heaviest _____

(f) that would be the lightest _____

(g) that has the most space/greatest volume _____

(h) that has the least space/smallest volume _____

(i) with the densest air _____

(j) with the least dense air _____

RATE MY UNDERSTANDING
Shade the face that shows your rating

2.4 Changes of state

Science understanding

FOUNDATION | **STANDARD** | ADVANCED

The three main states of matter—solid, liquid and gas are shown in Figure 2.4.1.

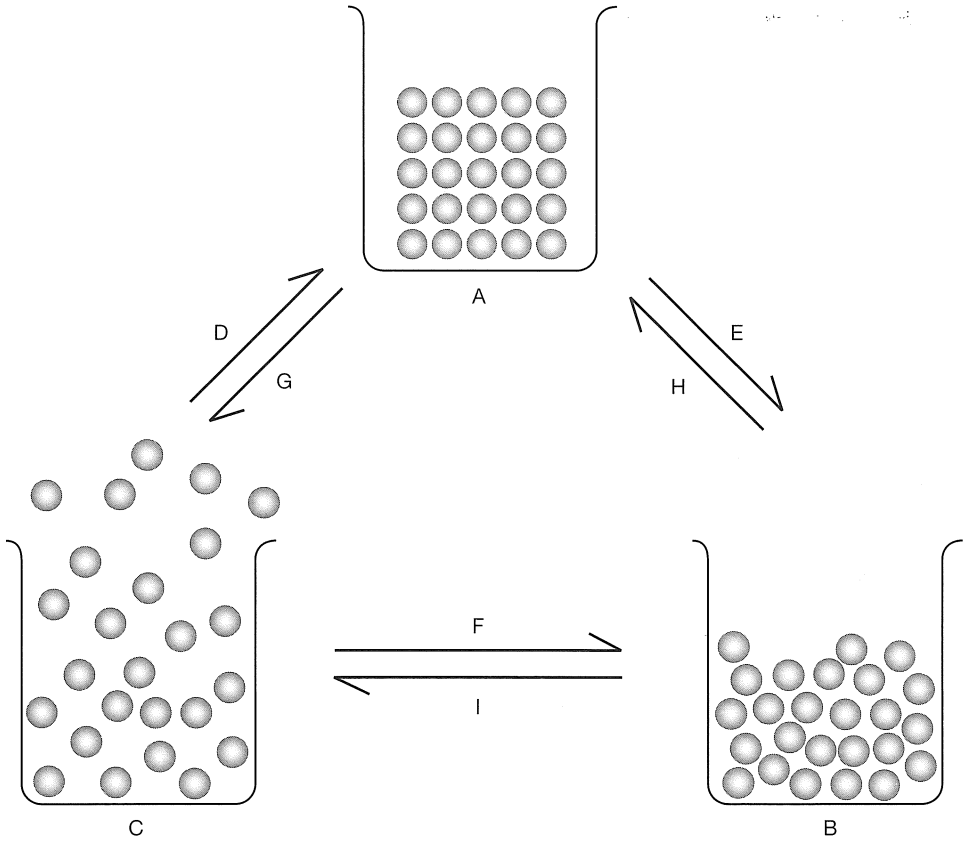


Figure 2.4.1

1 Identify which diagram (A, B or C) best represents a:

(a) solid _____ **(b)** liquid _____ **(c)** gas _____

2 Identify which arrows (D, E, F, G, H or I) represent the following changes of state:

(a) melting _____ **(d)** condensation _____

(b) freezing _____ **(e)** sublimation _____

(c) evaporation _____ **(f)** deposition _____

3 Explain why the melting point of a substance is the same as its freezing point.

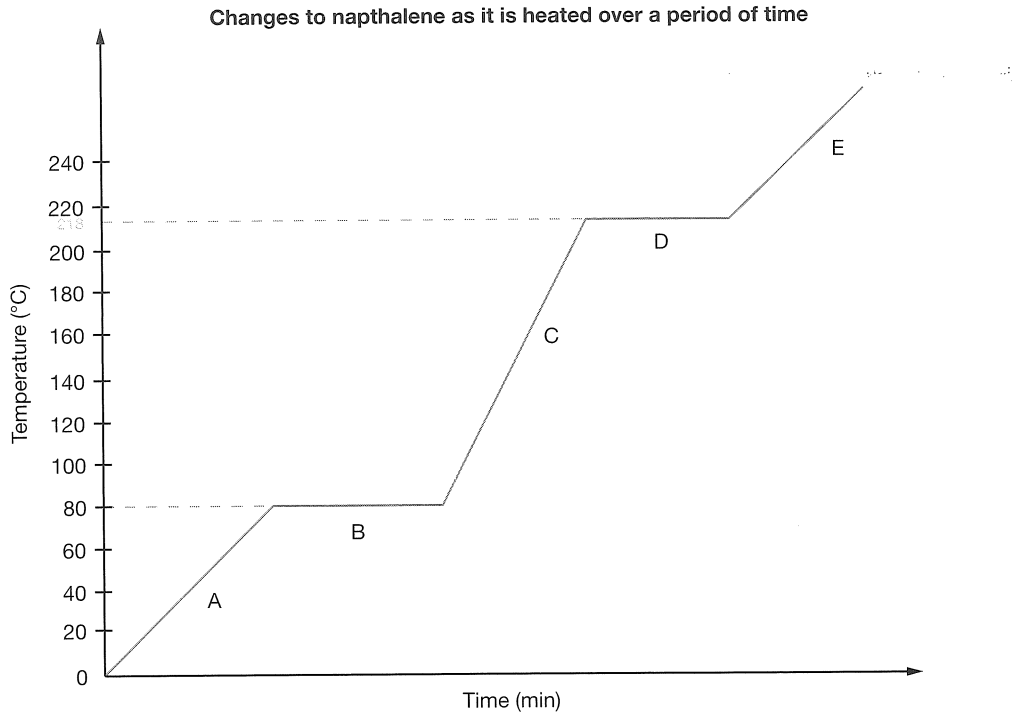
_____

_____

2.4 Changes of state

Naphthalene is a smelly chemical commonly used in mothballs. Some flakes of naphthalene were heated up until they melted then boiled. The graph below shows the important stages in this heating.

mothballs *(n)* small balls of chemicals, usually naphthalene, used to protect clothing from moths



4 Identify which section of the graph (A, B, C, D or E) best represents when naphthalene was:

(a) all gas _____

(b) in both gaseous and liquid states _____

(c) all liquid _____

(d) in both liquid and solid states _____

(e) all solid _____

5 Use the graph to predict the melting point of naphthalene. Which of the following options is correct?

A 0°C **C** 100°C

B 80°C **D** 218°C _____

6 Use the graph to predict the boiling point of naphthalene. Which of the following options is correct?

A 0°C **C** 100°C

B 80°C **D** 218°C _____

RATE MY UNDERSTANDING
Shade the face that shows your rating

2.5 Cooling curve

Science understanding

FOUNDATION	STANDARD	ADVANCED

Salty water was being heated on a hotplate. When it boiled it was then removed from the hotplate and placed in a freezer to cool. Its temperature was measured every minute. The measurements taken are shown in Table 2.5.1 below.

Table 2.5.1

Time (minutes)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Temperature (°C)	104	103	102	91	82	70	59	51	40	28	20	11	2	–1	–4	–4	–4

(1) **(a)** Construct a graph by plotting these values on the grid provided.

(b) Write a title for the graph.

(2) Read each statement and indicate whether it is true or false by circling the correct answer.

(a) After 2 minutes in the freezer, the salty water had a temperature of 104°C. True / False

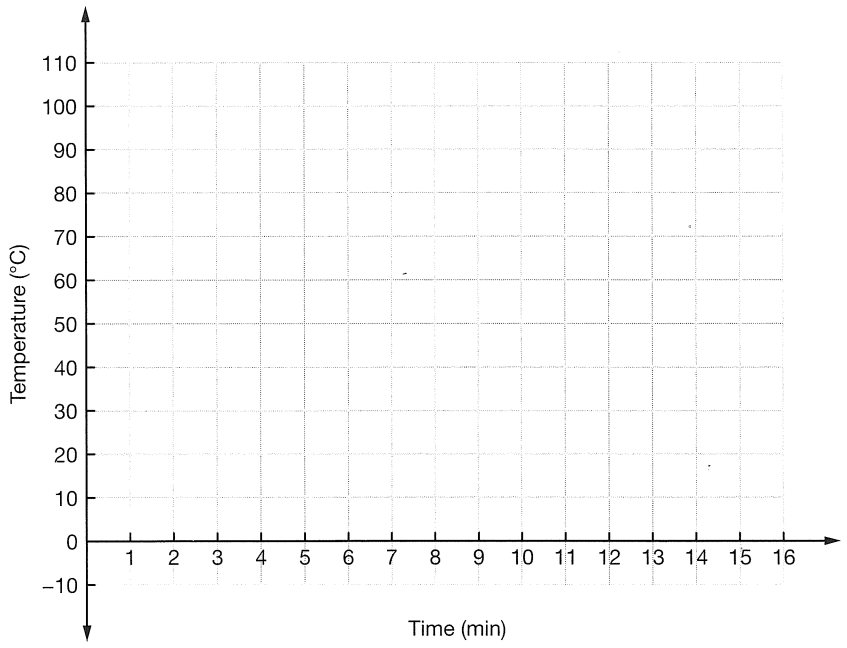
(b) The temperature reached 40°C after 8 minutes. True / False

(c) The temperature dropped by 12°C between the 8th and 9th minute. True / False

(d) Every minute the temperature dropped by 12°C. True / False

(e) The data was collected over 15 minutes. True / False

(f) Over the time data was collected, the temperature dropped by 108°C in total. True / False



(3) Use your graph to estimate:

(a) the boiling point of salty water _____

(b) the freezing point of salty water. _____

RATE MY UNDERSTANDING
Shade the face that shows your rating

2.6 Incomplete experiment report

Science understanding

FOUNDATION | **STANDARD** | ADVANCED

A cooling curve experiment was conducted to obtain the data shown in Worksheet 2.5, page 23. A complete investigation report was not provided.

Use the information in Worksheet 2.5 and the practical investigation report checklist in Figure 2.6.1 below to complete the following tasks.

controlled variable *(n)* factor that stays the same

dependent variable *(n)* what is being measure or tested

independent variable *(n)* what is being changed and how it is being changed to test the dependent variable

Practical investigation report checklist

Title

- ☐ what was investigated

Purpose

- ☐ the purpose describes what you wanted to show, prove or find out in an investigation
- ☐ can be a statement or a question
- ☐ one or two sentences
- ☐ often written as 'To investigate the effect of ... on ...'

Hypothesis

- ☐ a hypothesis is a prediction about the result of your investigation
- ☐ a short statement
- ☐ describes the different things you tested (these are called dependent and independent variables)
- ☐ not always included in a scientific report

Materials

- ☐ a list of all the important equipment, chemicals and materials that you used
- ☐ includes quantities of substances and sizes of equipment

Procedure

- ☐ the procedure or method is a detailed list of what you did in the experiment, in the exact order you did it
- ☐ written in short, numbered steps
- ☐ includes the quantities you used (e.g. 5 g, 2 spatula loads, 10 mL)
- ☐ can include diagrams of the experiment (2D scientific diagrams)

Results

- ☐ results are a record of all the observations and measurements you took during the investigation
- ☐ observations can be written and can include diagrams, photos and videos
- ☐ written observations are best presented in a table
- ☐ include any graphs or calculations

Review

- ☐ an analysis of your observations and measurements
- ☐ analyse any table, spreadsheet or graph you produced
- ☐ compare your findings with other groups or with information found from textbooks or the internet
- ☐ evaluate how you could make your investigation better
- ☐ construct a short conclusion that summarises what you found out in the experiment
- ☐ use your conclusion to evaluate how accurate your hypothesis was

Figure 2.6.1

1. On the checklist, tick all the parts of a report that are covered in Worksheet 2.5.

2. What do you think the aim of the experiment was?

_____

_____

3. Name the dependent variable.

_____

_____

2.6 Incomplete experiment report

4 Determine whether the data collected is qualitative or quantitative.

_____

> **qualitative data** *(n)* data collected as descriptions, e.g. hot
> **quantitative data** *(n)* data collected as numbers, e.g. 35°C

5 List the ways data is presented.

_____

_____

(6) A materials list is not included in Worksheet 2.5. Write the materials list exactly as it would appear on a practical investigation report.

_____

_____

_____

_____

_____

(7) Worksheet 2.5 does not include a description of the method used for the experiment, as it should appear on a practical investigation report. Write the method section of the practical investigation.

_____

_____

_____

_____

_____

8 This experiment was only carried out once. What could be done to ensure the results are fair?

_____

_____

RATE MY UNDERSTANDING
Shade the face that shows your rating ☺ 😐 ☹

4.4 Mixtures

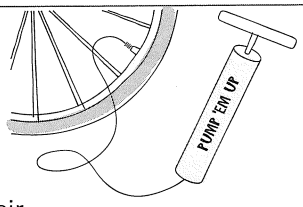
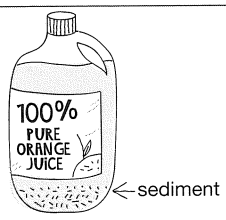
Science understanding

FOUNDATION | **STANDARD** | ADVANCED

homogenised (*adj*) milk that has been treated to prevent the solids and liquids from separating

sediment (*n*) solid particles that settle or rest at the bottom of a liquid

1 Classify the type of mixtures in the table below and identify the states of the substances in each.

Mixture	Type of mixture: solution, suspension?	Are its particles: solid in liquid, liquid in liquid, gas in liquid, gas in gas, or liquid in gas?
 milk		
 diluted cordial (ready to drink)		
 air		
 aerosol deodorant being sprayed		
 pure orange juice in bottle		
 medicine in bottle		

RATE MY UNDERSTANDING
Shade the face that shows your rating

4.2 Solubility and temperature

Science inquiry skills

FOUNDATION | STANDARD | ADVANCED

Processing & Analysing | Questioning & Predicting

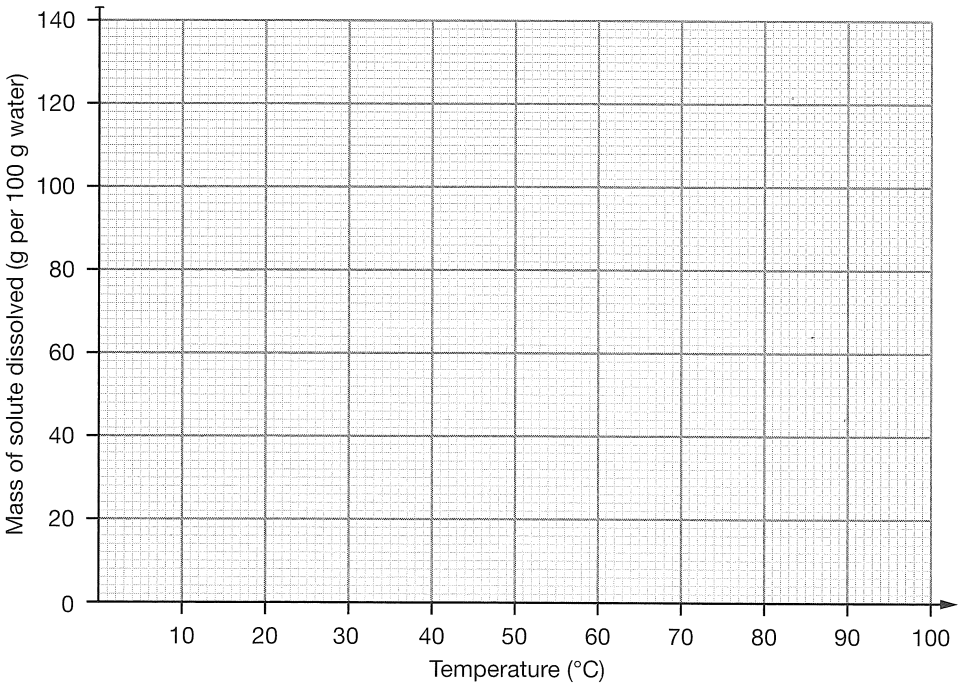
Maria and Tony did an experiment on solubility of substances at different temperatures. The aim was to find out if different substances have different solubilities when the temperature is changed. They tested how much copper sulfate would dissolve in water at different temperatures. They repeated the experiment with potassium sulfate. Table 4.2.1 shows their results.

Table 4.2.1 Results of solubilities of copper sulfate and potassium sulfate at different temperatures

Temperature (°C)	How much copper sulfate dissolves (g per 100 g water)	How much potassium sulfate dissolves (g per 100 g water)
0	22.3	7.4
10	27.2	9.3
20	32.3	11.1
30	37.8	13.0
40	44.8	14.8
50	52.8	16.5
60	62.5	18.2
70	73.4	19.8
80	87.5	21.4
90	105.4	22.9
100	124.9	24.1

(1) Plot both sets of data on the graph below. Label each line on the graph.

Results of solubilities of copper sulfate and potassium sulfate at different temperatures



2 In this experiment name the:

(a) solvent _____

(b) solutes. _____

3 Indicate whether the following statements are true or false by circling your answer:

(a) The copper sulfate solution is more concentrated at 80°C than at 0°C. True / False

(b) The copper sulfate solution is more dilute at 10°C than 50°C. True / False

(c) The potassium sulfate is more dilute at 60°C than 10°C. True / False

(d) The less copper sulfate in the water, the more concentrated the solution. True / False

4 State how much copper sulfate was dissolved in the water at the temperatures listed.

(a) 10°C _____

(b) 80°C _____

(c) 45°C _____

5 State how much potassium sulfate was dissolved in the water at temperatures listed.

(a) 30°C _____

(b) 25°C _____

(c) 100°C _____

6 Identify the effect that increasing the temperature had on the solubility of:

(a) copper sulfate

_____

(b) potassium sulfate.

_____

7 Which of the two chemicals, copper sulfate or potassium sulfate, was more soluble at any temperature? Give evidence from the table or graph to explain your answer.

_____

_____

8 Which of the two chemicals, copper sulfate or potassium sulfate, was more affected by increasing the temperature? Give evidence from the data to explain your answer.

_____

_____

RATE MY UNDERSTANDING
Shade the face that shows your rating ☺ 😐 ☹

4.3 Solubility of liquids

Science inquiry skills

FOUNDATION | STANDARD | ADVANCED

Processing & Analysing

David and Susan did an experiment mixing different substances. The aim was to test how well different substances mixed together. They placed 5 cm of a particular solvent in a test-tube, then added 1 cm of a solute. They then shook them together to see if the solute dissolved. Their results are shown in Table 4.3.1 below. A tick means they mixed well. A cross means they did not seem to mix and separated out into two layers.

Table 4.3.1 Results showing how well different liquids mixed together

Solvent (5 cm in test-tube)	Solute (1 cm added into test-tube of solvent)					
	Kerosene	Water	Methylated spirits	Cooking oil	Glycerol	Turpentine
Kerosene	no need to do	✘	✘	✔	✘	✔
Water	✘	no need to do	✔	✘	✔	✘
Methylated spirits	✘	✔	no need to do	✘	✔	✘
Cooking oil	✔	✘	✘	no need to do	✘	✔
Glycerol	✘	✔	✔	✘	no need to do	✘
Turpentine	✔	✘	✘	✔	✘	no need to do

1 Match the terms in the left-hand column with the definitions in the right-hand column by drawing lines between them:

Term
cooking oil
glycerol
kerosene
solute
methylated spirits
solvent
turpentine

Definition
a substance that dissolves another chemical
a bad smelling solvent used as a fuel and cleaner of paint
a substance that dissolves into another substance
a paint thinner
fats from plants and animals such as butter, coconut oil, sunflower oil, margarine
a liquid used as a fuel in industry and in homes
a colourless, odourless sugar alcohol used as a sweetener

(2) Identify the solutes that dissolved in water.

_____

(3) Identify the solutes that dissolved in oil.

_____

(4) Identify the solvents that dissolved oil.

_____

(5) Identify the solutes that did not dissolve in oil.

_____

(6) Identify two solvents for:

(a) methylated spirits _____

(b) kerosene _____

(c) turpentine. _____

(7) Identify a solute that dissolves in:

(a) glycerol and also in methylated spirits _____

(b) cooking oil and also in kerosene. _____

(8) State whether each of these mixtures would form a solution. Write 'yes' or 'no'.

(a) kerosene in cooking oil _____

(b) water in glycerol _____

(c) turpentine in kerosene _____

(d) turpentine in water _____

(9) David and Susan concluded that oil and water seem to be different types of solvents, and that there seemed to be two different types of substances—those that dissolved in water, and those that dissolved in cooking oil.

(a) List the substances that dissolved in water.

_____

(b) List the substances that dissolved in cooking oil.

_____

(c) State whether the substances that mixed well in water also mix well with each other. Explain your answer.

_____

_____

(10) Propose a way in which Susan and David could make substances that did not mix well with each other combine better.

_____

_____

RATE MY UNDERSTANDING
Shade the face that shows your rating ☺ 😐 ☹